Prodigy Anesthesia Quick Review

Evoked Potentials

What are evoked potentials?	Electrical potentials that are measured in response to some type of stimulus. The purpose is to improve or preserve neurologic structures that are at risk, and to also prevent damage that is irreversible. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 329.*
With regards to evoked potentials, what are three general rules to help guide a nurse anesthetist?	1. Lipophilic agents that interfere with neuronal membrane conduction also interfere with subcortical conduction. 2. At equipotent doses to inhalational agents, IV agents will affect evoked potentials to a lesser degree. 3. Anesthetic agents that interfere with the ECG will also affect evoked potentials. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
What is the purpose of SSEP's?	SSEP's monitor various neural structures along the central and peripheral somatosensory pathways. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
How is the stimulation for SSEP's created?	Through electrical stimulation of peripheral nerves *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
Do SSEP's provide real-time electrical waveforms?	No. They are processed as an average, using electrical filters in the removal of background noise. The interpretation of the compound action potential depends on the distance to the recording electrodes and the site of the stimulus. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
In what types of surgeries are SSEP's beneficial?	Since changes in SSEP's are sensitive to cerebral ischemia, they have many uses in vascular surgery, including carotid endarterectomy, and cerebral aneurysm surgery. SSEP's can also indicate high risk for neurologic injury during cross-clamping of the aorta. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
What anesthetic agents do not produce a change in latency or amplitude in regards to SSEP's?	Ketamine, etomidate, and opiates. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
If SSEP's are monitored alone, can a paralytic agent be used?	Yes. If SSEP's are being monitored, the use of a paralytic is accepted, and potentials are able to be measured. Motor responses will not be elicited, however. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*

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How are the lower extremities monitored during SSEP's?	Through the stimulation of the posterior tibial nerve, which is located between the Achilles tendon and medial malleolus of the ankle *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
Where is the stimulus placed to monitor SSEP's in the upper extremity?	At the location of the median nerve which can be found between the tendons of the flexor carpi radialis and the palmaris longus. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
Describe how VEPs are performed.	The patient is presented a visual stimulus for a select number of times, then the cerebral responses are dispayed after being amplified and averaged on a computer. There are two types of VEPs: patterned and unpatterned. Patterned tests are used on awake patients. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 332.*
What is the purpose of Visual Evoked Potentials (VEPs)?	They are used to monitor function of the visual pathway. This is composed of the retina, extending to the occipital cortex, including everything that exists in between, including the optic chiasm and the optic nerve. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 332.*
How are VEPs performed on anesthetised patients?	A stroboscopic flash stimulus is used, with recording electrodes placed on the patient's scalp. A flash VEP is derived from a flash that already has a determined strength, of certain light intensity against a dim background. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 332.*
What is the purpose of Brainstem Auditory Evoked Potentials (BAEP's)?	BAEP's are used in the monitoring of the entire auditory pathway, which allows for the monitoring of basic brainstem function. They are clinically useful because they are extremely resistant to abberation unless there is actual structural pathology in the brainstem. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
What is the stimulus used in the monitoring of BAEPs?	A standard broadband repeating click that is delivered through an earphone that is placed in the auditory canal. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 916.*
What substances can cause dampened sounds and delayed responses of BAEPs?	Saline, CSF, fluid, and soap. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*

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What factors can affect BAEPs?	Hypothermia can result in decreased latency and prolonged interpeak intervals, and low PCO2 exaggerates BAEPs, resulting in an increased latency. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 331.*
Do anesthetic agents affect BAEPs?	No. BAEPs are not significantly affected by benzodiazepines, ketamine, propofol, muscle relaxants, N2O, or barbiturates. Inhalation agents can affect BAEPs mildly, however the affect is proportional to the dose administered. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330-331.*
When should the transducer be placed when monitoring BAEPs?	After the head is positioned. This produces the best results, and also prevents abrasive injury to the ear canal. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 330.*
What is the purpose of motor evoked potentials?	Motor evoked potentials (MEPs) are useful in the monitoring of functional intergrity of motor tracts, namely, the corticospinal tract. The stimuli can be magnetic or electrical, with either the spinal cord or the motor cortex used as the site of stimulation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 331.*
What is required for the stimulation and monitoring of motor components of nerves?	An active stimulus to produce an action potential. Electromyography (EMG) is able to stimulate the motor nerve as well as monitor the given muscle groups being innervated. EMG also has the capability to be a passive listener to all other muscle groups. This passive monitoring of nerves is particularly useful to the surgeon because it allows for more awareness of what nerves are being stimulated during surgical manipulation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 331.*
MEPs using magnetic stimulation is contraindicated in what patients?	Patients with spinal or bladder stimulators, metallic foreign bodies, pacemakers, or a previous craniotomy *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 331.*
What type of stimulation (electrical or magnetic) is used more common in spine cases?	Electrical. Magnetic stimulation is extremely cumbersome in an operating room due to the interference of the magnetic field with operating room equipment, in addition to the position of the equipment and the size. Magnetic fields also generate a high-intensity noise, and produce heat. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 331.*

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